INTEGRATION

$$\int x^n dx = \frac{1}{n+1} x^{n+1} \cdot \frac{1}{\frac{d}{dx}(x)} + C,$$

which holds for all values of n except $n = -1$

$$\int a x^n dx = a \int x^n dx$$

$$= \frac{a}{n+1} x^{n+1} \cdot \frac{1}{\frac{d}{dx}(x)} + C$$

$$\int (ax+b)^n dx = \frac{1}{n+1} (ax+b)^{n+1} \cdot \frac{1}{\frac{d}{dx}(ax+b)} + C$$

$$= \frac{1}{n+1} (ax+b)^{n+1} \cdot \frac{1}{a} + C$$

$$= \frac{1}{a(n+1)} (ax+b)^{n+1} + C$$

Scanned by CamScanner

LEVEL 0

$$\int x\,dx = \frac{1}{2}x^2 + C$$

$$\int x^2\,dx = \frac{1}{3}x^3 + C$$

$$\int x^3\,dx = \frac{1}{4}x^4 \cdot \frac{1}{\frac{d}{dx}(x)} + C$$

$$= \frac{1}{4}x^4 + C$$

$$\int x^8\,dx = \frac{1}{9}x^9 \cdot \frac{1}{\frac{d}{dx}(x)} + C$$

$$= \frac{1}{9}x^9 + C$$

$$\int x^{50}\,dx = \frac{1}{51}x^{51} \cdot \frac{1}{\frac{d}{dx}(x)} + C$$

$$= \frac{1}{51}x^{51} + C$$

Scanned by CamScanner

Note:

$$\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

Ex 1

$$\int (5x^2 + 4x - 3)\,dx = \int 5x^2\,dx + \int 4x\,dx - \int 3\,dx$$

$$= 5\int x^2\,dx + 4\int x\,dx - 3\int dx$$

$$= \frac{5x^3}{3} + \frac{4x^2}{2} - 3x + C$$

$$= \frac{5}{3}x^3 + 2x^2 - 3x + C$$

Ex 2

$$\int (x^3 - 5x^2 + 7x - 11)\,dx = \int x^3\,dx - 5\int x^2\,dx + 7\int x\,dx - 11\int$$

$$= \frac{x^4}{4} - \frac{5}{3}x^3 + \frac{7}{2}x^2 - 11x + C$$

Ex 3

$$\int \left(\sqrt[3]{x} - \frac{1}{\sqrt[3]{x}}\right) = \int x^{\frac{1}{3}}\,dx - \int x^{-\frac{1}{3}}\,dx$$

$$= \frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} - \frac{x^{-\frac{1}{3}+1}}{-\frac{1}{3}+1} + C$$

$$= \frac{3}{4}x^{\frac{4}{3}} - \frac{3}{2}x^{\frac{2}{3}} + C$$

Scanned by CamScanner

INTEGRATING $\frac{1}{x}$

we apply the rule for integrating x^n is $\frac{1}{x}$ (ie x^{-1}), we find---

$$\int \frac{1}{x}\,dx = \int x^{-1}\,dx = \frac{x^{-1+1}}{-1+1} + C = \frac{x^0}{0} + C$$

which involves a division by zero.

This shows that the rule

$$\int x^n\,dx = \frac{1}{n+1}x^{n+1} \cdot \frac{1}{\frac{d}{dx}(x)} + C$$

does not work in the case where $n = -1$

THEOREM

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$$

Thus $\int x^{-1}\,dx = \int \frac{1}{x}\,dx$

$$= \ln|x| + C$$

Scanned by CamScanner

NATURAL LOGARITHMIC FUNCTIONS

Example 1

$$\int \frac{dx}{ax+b} = \frac{1}{a} \ln|ax+b| + C$$

Example 2

$$\int \frac{1}{x} dx = \ln|x| + C$$

Example 3

$$\int \frac{1}{3x} dx = \frac{1}{3}\int \frac{1}{x} dx$$

$$= \frac{1}{a}\frac{1}{3} \ln|x| + C$$

Example 4

$$\int \frac{1}{7x-4} dx = ?$$

Let $u = 7x-4$

$du = 7dx$

$$\int \frac{1}{7x-4} dx = \int \frac{1}{u} \cdot \frac{du}{7} = \frac{1}{7}\int \frac{1}{u} du = \frac{1}{7} l$$

$$= \frac{1}{7} \ln|7x-4| + C$$

Example 5

$$\int \frac{x^3}{x^4-1} dx = ?$$

Let $u = x^4-1$

$du = 4x^3 dx \implies x^3 dx = \frac{du}{4}$

Scanned by CamScanner

$$\rightarrow \int \frac{x^3}{x^4-1}\,dx = \int \frac{1}{u}\cdot\frac{du}{4} = \frac{1}{4}\int \frac{du}{u} = \frac{1}{4}\ln|u| + c$$

$$= \frac{1}{4}\ln|x^4-1| + C$$

Example 6

$$\int \cot x\,dx = ?$$

Let $u = \sin x$, $du = \cos x\,dx$

$$\Rightarrow \int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx = \int \frac{1}{u}\,du = \ln|u| + C$$

$$= \ln|\sin x| + C$$

Example 7

$$\int \frac{1}{x\ln x}\,dx = ?$$

Let $u = \ln x$;

$$du = \frac{1}{x}\,dx$$

$$\Rightarrow \int \frac{1}{x\ln x}\,dx = \int \frac{1}{u}\,du = \ln|u| + C = \ln(\ln|x|) + C$$

Example 8

$$\int \frac{\cos 2x}{1-\sin 2x}\,dx = ?$$

Let $u = 1 - \sin 2x$

$$du = -2\cos 2x\,dx$$

$$\Rightarrow \int \frac{\cos 2x}{1-\sin 2x}\,dx = -\frac{1}{2}\int \frac{1}{u}\,du = -\frac{1}{2}\ln|u| + C$$

$$= -\frac{1}{2}\ln|1-\sin 2x| + C$$

Scanned by CamScanner

(b)(i) $\int \frac{1}{2}x^{-\frac{1}{2}}dx$ (ii) $\int \frac{1}{2\sqrt{2x^3}}dx$ (iii) $\int \frac{dx}{\sqrt[3]{x}}$

2.(a)(i) $\int\left(\frac{1}{x^3}-\frac{1}{x^2}+\frac{1}{x}-1\right)dx$ (ii) $\int\left(2-\frac{1}{3}x^2-\frac{1}{2\sqrt{x}}\right)$

(b)(i) $\int\{(x+3)(x-3)\}dx$ (ii) $\int\left(x^{\frac{1}{3}}+1+x^{-\frac{1}{3}}\right)dx$.

DEFINITE INTEGRALS

Example 1

Evaluate $\int_1^2 \frac{x-1}{x^2-2x+2}dx$

Soln:

$$\int_1^2 \frac{x-1}{x^2-2x+2}dx = \frac{1}{2}\int_1^2 \frac{2x-2}{x^2-2x+2}dx$$

$$= \frac{1}{2}\ln|x^2-2x+2|\Big]_1^2$$

$$= \frac{1}{2}(\ln 2 - \ln 1)$$

$$= \frac{1}{2}\ln 2$$

Example 2

Find $\int_1^2 \frac{x}{3x^2-2}dx$

Soln:

$$\int_1^2 \frac{x}{3x^2-2}dx = \frac{1}{6}\int_1^2 \frac{6x}{3x^2-2}dx = \frac{1}{6}\ln|3x^2-2|_1^2$$

$$= \frac{1}{6}(\ln 10 - \ln 1)$$

$$= \frac{1}{6}\ln 10$$

Scanned by CamScanner

Example 3

Evaluate $\int_{\pi/6}^{\pi/2} \frac{\cos x}{2-\sin x}\,dx$

Soln:

$$\int_{\pi/6}^{\pi/2} \frac{\cos x}{2-\sin x}\,dx = -\int_{\pi/6}^{\pi/2} \frac{-\cos x}{2-\sin x}\,dx = -\ln|2-\sin x|\Big]_{\pi/6}^{\pi/2}$$

$$= -(\ln 1 - \ln \tfrac{3}{2})$$

$$= \ln \tfrac{3}{2} = \ln 3 - \ln 2$$

EXPONENTIAL FUNCTIONS

$$\int e^x dx = e^x + C$$

$$\int a e^x dx = a e^x + C$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

Example

(i) $\int e^{3x} dx = \frac{1}{3} e^{3x} + C$

(ii) $\int 2e^{5x} dx = 2\int e^{5x} dx$

$= \frac{2}{5} e^{5x} + C$

Scanned by CamScanner

INTEGRALS OF STANDARD FORMS

There are a number of integrals known STANDARD FORMS. They are derivatives of which could be applied or used as such.

ALGEBRAIC FUNCTIONS

1. $\int x^n dx = \frac{1}{n+1} x^{n+1} + C$

2. $\int \frac{dx}{x} = \ln|x| + C$

3. $\int a^x dx = \frac{1}{\ln a} a^x + C$

4. $\int e^x dx = e^x + C$

TRIGONOMETRIC FUNCTIONS

1. $\int \sin x \, dx = -\cos x + C$

2. $\int \cos x \, dx = \sin x + C$

3. $\int \tan x \, dx = -\ln|\cos x| + C = \ln|se$

4. $\int \cot x \, dx = \ln|\sin x| + C$

Note:

The derivatives of sec x and cosec x nam
sec x tan x and $-\text{cosec}\, x \cot x$, do not give

Scanned by CamScanner

standard forms, but to products of them. The integrals of $\sec x$ and $\operatorname{cosec} x$ do not arise by direct differentiation.

5. $\int \sin ax\, dx = -\frac{1}{a}\cos x + C$

6. $\int \sin(ax+b)\, dx = -\frac{1}{a}\cos(ax+b) + C$

7. $\int \cos ax\, dx = \frac{1}{a}\sin ax + C$

8 $\int \cos(ax+b)dx = \frac{1}{a}\sin(ax+b) + C$

9. $\int \tan ax\, dx = \frac{1}{a}\ln|\sec ax| + C$

INVERSE TRIGONOMETRIC FUNCTIONS

1. $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}x + C$

2. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$

3. $-\int \frac{dx}{\sqrt{a^2-x^2}} = \cos^{-1}\frac{x}{a} + C$

4. $-\int \frac{dx}{\sqrt{1-x^2}} = \cos^{-1}x + C$

Scanned by CamScanner

5. $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$

6. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

7. $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{x}{a}$

ADDITIONAL STANDARD INTEGRALS

The following are additional standard forms which are useful for integration. Some of them are results of differentiating standard forms

1. $\int \sec x \tan x \, dx = \sec x + C$
2. $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C$
3. $\int \operatorname{cosec}^2 x \, dx = -\cot x + C$
4. $\int \sec^2 x \, dx = \tan x + C$

HYPERBOLIC FUNCTIONS

1. $\int \sinh x \, dx = \cosh x + C$
2. $\int \cosh x \, dx = \sinh x + C$
3. $\int \tanh x \, dx = \ln \cosh x + C$
4. $\int \coth x \, dx = \ln|\sinh x| + C$

Scanned by CamScanner

INVERSE HYPERBOLIC FUNCTIONS

1. $\int \frac{dx}{\sqrt{x^2+a^2}} = \sinh^{-1}\frac{x}{a} + c$

2. $\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\frac{x}{a} + c$

3. $\int \frac{dx}{a^2-x^2} = \frac{1}{a}\tanh^{-1}\frac{x}{a} + c$

Exercises

Find the following integrals

1. (i) $\int 3e^{2x}dx$ (ii) $\int e^{3x-1}dx$ (iii) $\int e^{\frac{x}{a}}dx$

2. (i) $\int \frac{x^2-x+1}{x^3}dx$ (ii) $\int \sqrt{ax+b}\,dx$ (iii) $\int \frac{dx}{\sqrt{ax+b}}$

3. (i) $\int \frac{x\,dx}{x^2-1}$ (ii) $\int \frac{dx}{\sqrt{1-x}}$ (iii) $\int \sqrt{1+\frac{x}{2}}\,dx$

4. (i) $\int x(1+x)(1+x^2)dx$ (ii) $\int \frac{\sin ax}{1+\cos ax}dx$

5. (i) $\int \frac{e^{3x}}{e^{3x}+6}dx$ (ii) $\int \frac{1+\cos 2x}{2x+\sin 2x}dx$

6. (i) $\int \sin\frac{1}{3}x\,dx$ (ii) $\int \sin(\alpha-3x)dx$ (iii) $\int \cos(2x+\alpha)dx$

7. (i) $\int \sin 3x\,dx$ (ii) $\int \cos 5x\,dx$

Scanned by CamScanner

USING SUBSTITUTION

Example 1

$\int (3x-5)^{12} dx = ?$

Let $u = 3x-5$

$du = 3dx \Rightarrow dx = \frac{du}{3}$

$\Rightarrow \int (3x-5)^{12} dx = \int u^{12} \cdot \frac{1}{3} du = \frac{1}{3}\int u^{12} du$

$= \left(\frac{1}{3}\right)\left(\frac{1}{13}\right) u^{13} + C = \frac{1}{39} u^{13} + C$

$= \frac{1}{39}(3x-5)^{13} + C$

Example 2

$\int \sqrt{7x+4}\, dx = ?$

Let $u = 7x+4$

$du = 7dx \Rightarrow dx = \frac{du}{7}$

$\Rightarrow \int \sqrt{7x+4}\, dx = \int \sqrt{u} \cdot \frac{du}{7} = \frac{1}{7}\int u^{\frac{1}{2}} du$

$= \left(\frac{1}{7}\right)\left(\frac{2}{3}\right) u^{\frac{3}{2}} + C$

$= \frac{2}{21} u^{\frac{3}{2}} + C$

$= \frac{2}{21}(7x+4)^{3/2} + C$

Example 3

$\int \frac{1}{\sqrt{x-1}} dx = ?$

Let $u = x-1$

$du = dx$

Scanned by CamScanner

Example 8

$\int (x^4+1)^{\frac{1}{3}} x^7\,dx = ?$

Let $u = x^4+1 \Rightarrow x^4 = u-1$

$du = 4x^3\,dx \Rightarrow x^3\,dx = \frac{du}{4}$

Also $x^7\,dx = x^4 \cdot x^3\,dx = (u-1)\cdot\frac{du}{4}$

$\Rightarrow \int (x^4+1)^{\frac{1}{3}} x^7\,dx = \int u^{\frac{1}{3}}(u-1)\cdot\frac{du}{4}$

$= \int \frac{1}{4}\left(u^{\frac{4}{3}} - u^{\frac{1}{3}}\right)du$

$= \frac{1}{4}\left(\frac{3}{7}u^{\frac{7}{3}} - \frac{3}{4}u^{\frac{4}{3}}\right) + C$

$= \frac{3}{4}u^{\frac{4}{3}}\left(\frac{1}{7}u - \frac{1}{4}\right) + C$

$= \frac{3}{4}(x^4+1)^{\frac{4}{3}}\left[\frac{1}{7}(x^4+1) - \frac{1}{4}\right] + C$

$= \frac{3}{4}(x^4+1)^{\frac{4}{3}}\left[\frac{4(x^4+1)-7}{28}\right] + C$

$= \frac{3}{4}(x^4+1)^{\frac{4}{3}}\left(\frac{4x^4+4-7}{28}\right)$

$= \frac{3}{112}(x^4+1)^{\frac{4}{3}}(4x^4-3) + C$

Example 9

$\int \frac{x}{\sqrt{1+5x^2}}\,dx = ?$

Soln: Let $u = 1+5x^2$

$du = 10x\,dx \Rightarrow x\,dx = \frac{du}{10}$

Scanned by CamScanner

$$\Rightarrow \int \frac{x}{\sqrt{1+5x^2}}\,dx = \int \frac{1}{\sqrt{u}} \cdot \frac{du}{10} = \frac{1}{10}\int u^{-\frac{1}{2}}\,du$$

$$= \frac{1}{10} \cdot 2u^{\frac{1}{2}} + C$$

$$= \frac{1}{5}\sqrt{1+5x^2} + C$$

Example 10

$\int x\sqrt{ax+b}\,dx$, when $a \neq 0$??

Let $u = ax+b \Rightarrow x = \frac{u-b}{a}$

$du = a\,dx \Rightarrow dx = \frac{du}{a}$

$$\Rightarrow \int x\sqrt{ax+b}\,dx = \int \frac{u-b}{a}\sqrt{u} \cdot \frac{du}{a}$$

$$= \frac{1}{a^2}\int (u^{\frac{3}{2}} - bu^{\frac{1}{2}})\,du$$

$$= \frac{1}{a^2}\left(\frac{2}{5}u^{\frac{5}{2}} - \frac{2b}{3}u^{\frac{3}{2}}\right) + C$$

$$= \frac{2}{a^2}u^{\frac{3}{2}}\left(\frac{1}{5}u - \frac{b}{3}\right) + C$$

$$= \frac{2}{a^2}(\sqrt{ax+b})^3\left[\frac{1}{5}(ax+b) - \frac{b}{3}\right] + C$$

$$= \frac{2}{a^2}(\sqrt{ax+b})^3\left(\frac{a}{5}x - \frac{2}{15}b\right) + C$$

$$= \frac{2}{15a^2}(\sqrt{ax+b})^3(3ax - 2b)$$

Scanned by CamScanner

USING SUBSTITUTIONS FOR TRIGONOMETRIC FUNCTIONS

Example 1

$\int \sin(3x-1)dx = ?$

Let $u = 3x-1$

$du = 3dx \Rightarrow dx = \frac{du}{3}$

$\Rightarrow \int \sin(3x-1)dx = \int \sin u \cdot \frac{du}{3} = \frac{1}{3}\int \sin u\, du$

$= \frac{1}{3}(-\cos u) + C$

$= -\frac{1}{3}\cos(3x-1) + C$

Example 2

$\int \frac{\cos\sqrt{x}}{\sqrt{x}} dx = ?$

Let $u = \sqrt{x} = x^{\frac{1}{2}}$

$du = \frac{1}{2}\left(\frac{1}{\sqrt{x}}\right)dx$

$\int \frac{\cos\sqrt{x}}{\sqrt{x}} dx = \int \cos u \cdot 2\,du = 2\int \cos u\, du$

$= 2\sin u + C$

$= 2\sin(\sqrt{x}) + C$

Example 3

$\int \sec^2\left(\frac{x}{2}\right)dx$

Let $u = \frac{x}{2}$

$du = \frac{dx}{2} \Rightarrow dx = 2\,du$

Scanned by CamScanner

$$\Rightarrow \int \sec^2\left(\frac{x}{2}\right)dx = \int \sec^2 u \cdot 2\,du = 2\int \sec^2 u\,du$$

$$= 2\tan u + c$$

$$= 2\tan\left(\frac{x}{2}\right) + c$$

Example 4

$$\int \frac{\cos 3x}{\sin^2 3x}\,dx = ?$$

Let $u = \sin 3x$

$$du = 3\cos 3x\,dx \Rightarrow \frac{du}{3} = \cos 3x\,dx$$

$$\Rightarrow \int \frac{\cos 3x}{\sin^2 3x}\,dx = \int \frac{1}{u^2}\cdot\frac{du}{3} = \frac{1}{3}\int u^{-2}\,du$$

$$= \frac{1}{3}\left(-u^{-1}\right) + c$$

$$= -\frac{1}{3u} + c$$

$$= -\frac{1}{3\sin 3x} + c = -\frac{1}{3}\csc 3x + c$$

Example 5

$$\int \sin^4 x \cos x\,dx = ?$$

Let $u = \sin x$

$$du = \cos x\,dx$$

$$\Rightarrow \int \sin^4 x \cos x\,dx = \int u^4\,du = \frac{1}{5}u^5 + c$$

$$= \frac{1}{5}\sin^5 x + c$$

Scanned by CamScanner

Example 12

$\int \sec^5 x \tan x \, dx = ?$

Let $u = \sec x$

$du = \sec x \tan x \, dx$

$\Rightarrow \int \sec^5 x \tan x \, dx = \int u^4 du = \frac{1}{5} u^5 + C$

$= \frac{1}{5} \sec^5 x + C$

Example 13

$\int \left(\frac{\sec x}{1+\tan x} \right)^2 dx = ?$

Let $u = 1 + \tan x$

$du = \sec^2 x \, dx$

$\Rightarrow \int \left(\frac{\sec x}{1+\tan x} \right)^2 dx = \int u^{-2} du = -u^{-1} + C$

$= -\frac{1}{1+\tan x} + C$

RATIONALIZATION

Example 1

$\int \frac{dx}{1+\cos x} = ?$

$\frac{1}{1+\cos x} = \frac{1}{1+\cos x} \cdot \frac{1-\cos x}{1-\cos x} = \frac{1-\cos x}{\sin^2 x}$

$= \operatorname{cosec}^2 x - \cot x \operatorname{cosec} x$

$\int \frac{dx}{1+\cos x} = \int (\csc^2 x - \cot x \csc x) dx$

$= -\cot x + \csc x + C$

Scanned by CamScanner

Example 2

$$\int \frac{dx}{1+\sin x} = ?$$

$$\frac{1}{1+\sin x} = \frac{1}{1+\sin x} \cdot \frac{1-\sin x}{1-\sin x} = \frac{1-\sin x}{1-\sin^2 x} = \frac{1}{\cos^2 x} - \tan x \sec x = \sec^2 x - \sec x \tan x$$

$$\Rightarrow \int \frac{dx}{1+\sin x} = \int (\sec^2 x - \sec x \tan x)\, dx$$

$$= \tan x - \sec x + C$$

USING IDENTITIES AND DEFINITE LIMITS

$$\cos^2 x = \frac{(1+\cos 2x)}{2} \qquad \sin 2x = 2 \sin x$$

$$\sin^2 x = \frac{(1-\cos 2x)}{2} \qquad \sin^2 x + \cos^2 x = 1$$

Example 1

$$\int \cos^2 x\, dx = ?$$

$$\int \cos^2 x\, dx = \frac{1}{2}\int (1+\cos 2x)\, dx$$

$$= \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) + C$$

$$= \frac{1}{2}(x + \sin x \cdot \cos x) + C$$

Example 2

$$\int \sin^2 x\, dx = ?$$

$$\int \sin^2 x\, dx = \int (1-\cos^2 x)\, dx = x - \frac{1}{2}(x + \sin x \cos x)$$

$$= \frac{1}{2}(x - \sin x \cos x) + C$$

Scanned by CamScanner

Example 6

$$\int_0^{\pi/2} \cos x \sin x \, dx = ?$$

$$\int_0^{\pi/2} \cos x \sin x \, dx = \frac{1}{2}\sin^2 x \Big|_0^{\pi/2} = \frac{1}{2}\left[\sin^2\left(\frac{\pi}{2}\right) - \sin^2 0\right]$$

$$= \frac{1}{2}$$

Example 7

$$\int_0^{\pi/4} \tan x \sec^2 x \, dx = ?$$

$$\int_0^{\pi/4} \tan x \sec^2 x \, dx = \frac{1}{2}\tan^2 x \Big|_0^{\pi/4} = \frac{1}{2}\left[\tan^2\frac{\pi}{4} - \tan^2 0\right]$$

$$= \frac{1}{2}(1-0)$$

$$= \frac{1}{2}$$

Example 8

$$\int_0^{\pi/2} \sqrt{\sin x + 1}\, \cos x \, dx = ?$$

Let $u = \sin x + 1$; Solve !!!

$$\int_0^{\pi/2} \sqrt{\sin x + 1}\, \cos x \, dx = \int_0^{\pi/2} (\sin x + 1)^{1/2} \cos x \, dx = \frac{2}{3}(\sin x + 1)^{3/2}$$

$$= \frac{2}{3}\left\{\left[\sin\left(\frac{\pi}{2}\right) + 1\right]^{3/2} - (\sin(0) + 1)^{3/2}\right\}$$

$$= \frac{2}{3}(2\sqrt{2} - 1)$$

Scanned by CamScanner

Exx 3

$$\int \frac{x^3}{\sqrt{x^2+25}}\,dx = ?$$

Let $u = x^2+25 \Rightarrow x^2 = u-25$

$du = 2x\,dx \Rightarrow \frac{du}{2} = x\,dx$

$$\int \frac{x^3}{\sqrt{x^2+25}}\,dx = \int \frac{u-25}{\sqrt{u}} \cdot \frac{1}{2}\,du$$

$$= \frac{1}{2}\int \left(u^{\frac{1}{2}} - 25u^{-\frac{1}{2}}\right)du$$

$$= \frac{1}{2}\left(\frac{2}{3}u^{\frac{3}{2}} - 50u^{\frac{1}{2}}\right) + C$$

$$= \frac{1}{2}u^{\frac{1}{2}}(u-75) + C$$

$$= \frac{1}{2}\sqrt{x^2+25}(x^2-50) + C$$

Exx 4

$$\int (2x^3+x)(x^4+x^2+1)^{49}\,dx = ?$$

Let $u = x^4+x^2+1$

$du = (4x^3+2x)dx = 2(2x^3+x)dx$

$$\Rightarrow \int (2x^3+x)(x^4+x^2+1)^{49}\,dx = \int u^{49}\,\frac{du}{2}$$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{50}\right)u^{50} + C$$

$$= \frac{1}{100}u^{50} + C$$

$$= \frac{1}{100}(x^4+x^2+1)^{50} + C$$

Scanned by CamScanner

SOME TRIGONOMETRIC SUBSTITUTIONS

Example 1

$$\int \frac{dx}{\sqrt{a^2-x^2}} = ?$$

Let $x = a\sin\theta \Rightarrow \sin\theta = \frac{x}{a} \Rightarrow \theta = \sin^{-1}\frac{x}{a}$

$dx = a\cos\theta\, d\theta$

$$\Rightarrow \int \frac{dx}{\sqrt{a^2-x^2}} = \int \frac{a\cos\theta\, d\theta}{\sqrt{a^2-a^2\sin^2\theta}} = \int \frac{a\cos\theta\, d\theta}{\sqrt{a^2(1-\sin^2\theta)}}$$

$$= \int \frac{a\cos\theta\, d\theta}{\sqrt{a^2\cos^2\theta}}$$

$$= \int \frac{a\cos\theta\, d\theta}{a\cos\theta} = \int d\theta = \theta + C$$

$$= \sin^{-1}\frac{x}{a} + C$$

Example 2

$$\int \frac{dx}{a^2+x^2} = ?$$

NB. The form of the integral suggests the substitution $x = a\tan\theta$

Let $x = a\tan\theta \Rightarrow \tan\theta = \frac{x}{a} \Rightarrow \theta = \tan^{-1}$

$dx = a\sec^2\theta\, d\theta$

$$\Rightarrow \int \frac{dx}{a^2+x^2} = \int \frac{a\sec^2\theta\, d\theta}{a^2+a^2\tan^2\theta} = \int \frac{a\sec^2\theta\, d\theta}{a^2(1+\tan^2\theta)}$$

$$= \int \frac{a\sec^2\theta\, d\theta}{a^2\sec^2\theta}$$

Scanned by CamScanner

Example 4

$$\int \frac{dx}{x^2\sqrt{1+x^2}} = ?$$

Let $x = \tan\theta$; Also $\sqrt{1+x^2} = \sec\theta$

$dx = \sec^2\theta\, d\theta$

$$\Rightarrow \int \frac{dx}{x^2\sqrt{1+x^2}} = \int \frac{\sec^2\theta\, d\theta}{\tan^2\theta\sqrt{1+\tan^2\theta}} = \int \frac{\sec^2\theta\, d\theta}{\tan^2\theta \sec\theta}$$

$$= \int \frac{\sec\theta\, d\theta}{\tan^2\theta} = \int \frac{1}{\cos\theta} \cdot \frac{\cos^2\theta\, d\theta}{\sin^2\theta}$$

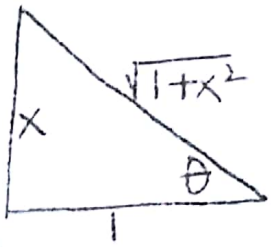
$$= \int \frac{\cos\theta}{\sin^2\theta}\, d\theta$$

Again let $u = \sin\theta$

$du = \cos\theta\, d\theta$

$$\Rightarrow \int \frac{\cos\theta}{\sin^2\theta}\, d\theta = \int \frac{du}{u^2} = \int u^{-2}\, du = -\frac{1}{u} + C$$

$$= -\frac{1}{\sin\theta} + C$$



$\sin\theta = \frac{x}{\sqrt{1+x^2}}$

But $\tan\theta = x \Rightarrow$

$$\Rightarrow \sin\theta = \frac{x}{\sqrt{1+x^2}}$$

$$= -\frac{\sqrt{1+x^2}}{x} + C$$

$$\int \frac{dx}{x^2\sqrt{1+x^2}} = -\frac{\sqrt{1+x^2}}{x} + C$$

Scanned by CamScanner

$$= \frac{1}{a}\int d\theta \quad = \frac{1}{a}\theta + C$$

$$= \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

Example 3

$$\int \sqrt{16-9x^2}\,dx = ?$$

Let $x = \frac{4}{3}\sin\theta$, $\Rightarrow \sin\theta = \frac{3}{4}x$; $\Rightarrow \theta = \sin^{-1}$

$dx = \frac{4}{3}\cos\theta\,d\theta$ [Also $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-(\frac{3}{4}x)^2} = \frac{1}{4}$

$$\Rightarrow \int \sqrt{16-9x^2}\,dx = \int \sqrt{16-9\left(\frac{16}{9}\sin^2\theta\right)}\cdot\frac{4}{3}\cos\theta\,d\theta$$

$$= \int \sqrt{16-16\sin^2\theta}\cdot\frac{4}{3}\cos\theta\,d\theta \quad [\sqrt{1-\sin^2\theta}$$

$$= \int 4\cos\theta\,\frac{4}{3}\cos\theta\,d\theta$$

$$= \frac{16}{3}\int \cos^2\theta\,d\theta$$

$$= \frac{16}{3}\int \frac{1+\cos 2\theta}{2}\,d\theta$$

$$= \frac{8}{3}\left(\theta + \frac{1}{2}\sin 2\theta\right) + C$$

$$= \frac{8}{3}\left(\sin^{-1}\frac{3}{4}x + \sin\theta\cos\theta\right) + C$$

$$= \frac{8}{3}\left(\sin^{-1}\frac{3}{4}x + \left(\frac{3x}{4}\right)\left(\frac{1}{4}\sqrt{16-9x^2}\right)\right) + C$$

$$= \frac{8}{3}\sin^{-1}\frac{3}{4}x + \frac{1}{2}x\sqrt{16-9x^2} + C$$

Scanned by CamScanner

INTEGRATION BY PARTS

This method for integration comes from the rule for differentiating a product using the product rule :

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v.\frac{du}{dx}$$, where u and v are functions of x

Integrating throughout with respect to x, we have

$$uv = \int u\frac{dv}{dx}dx + \int v\frac{du}{dx}dx$$

Rearranging, we have

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

Since u and v are functions of x, we can write the relation in the form

$$\int u\,dv = uv - \int v\,du$$

OR

$$\int f(x)\,g'(x)dx = f(x)\,g(x) - \int f'(x)\,g(x)dx$$

where $u = f(x)$ and $v = g(x)$

Example 1

$\int x \sin x\,dx = ?$

Let $u = x$; $dv = \sin x\,dx$

$du = dx$ and $v = \int \sin x\,dx$

$= -\cos x$.

[We ignore the constant of integration for the moment]

Scanned by CamScanner

$\Rightarrow \int x \sin x\,dx = -x\cos x + \int \cos x\,dx$

$= -x\cos x + \sin x + C$

NB It is at the final stage that we put in the constant of integration

Example 2

$\int x \cos x\,dx = ?$

Let $u = x$ and $dv = \cos x\,dx$

$du = dx$ $\quad v = \int \cos x\,dx = \sin x$

$\Rightarrow \int x\cos x\,dx = x\sin x + \int \sin x\,dx$

$= x\sin x + \cos x + C$

Example 3

$\int x^2 \sin x\,dx = ?$

Let $u = x^2$ and $dv = \sin x\,dx$

$du = 2x\,dx$ and $v = \int \sin x\,dx$

$= -\cos x$

Substituting in $\int u\,dv = uv - \int v\,du$, we have

$\int x^2 \sin x\,dx = x^2(-\cos x) - \int (-\cos x)(2x\,dx)$

$= -x^2\cos x + 2\int x\cos x\,dx$

Applying integration by parts again to the latter integral we have:

$\int x\cos x\,dx = ?$

Let $u = x$ and $dv = \cos x\,dx$

$du = dx$ and $v = \int \cos x\,dx = \sin x$

$\Rightarrow \int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x$

Scanned by CamScanner

$$\Rightarrow \int x^2 \sin x\,dx = -x^2\cos x + 2(x\sin x + \cos x) + C$$
$$= -x^2\cos x + 2x\sin x + 2\cos x + C$$

Example 4

$$\int \ln x\,dx = ?$$

Since $\ln x$ produces a simple expression when we differentiate it, we use the substitution for $\ln x$

Let $u = \ln x$ and $dv = (1)dx = dx$

$\Rightarrow du = \frac{1}{x}dx$ and $v = \int dx = x$

Using $\int u\,dv = uv - \int v\,du$, we have

$$\int \ln x\,dx = (\ln|x|)(x) - \int x\left(\frac{1}{x}\right)dx$$
$$= x\ln x - \int dx$$
$$= x\ln x - x + C$$

Example 5

$$\int x^2 e^{-x}\,dx = ?$$

We use integration by parts: $\int u\,dv = uv - \int v\,du$

In this case,

Let $u = x^2$, $dv = e^{-x}dx$

$\Rightarrow du = 2x\,dx$; $v = \int e^{-x}dx = -e^{-x}$

Thus $\int x^2 e^{-x}dx = -x^2e^{-x} + 2\int x e^{-x}dx$

To calculate the latter integral, we use another

Scanned by CamScanner

integration by parts for $\int x e^{-x} dx$

Let $u = x$; $dv = e^{-x} dx$

$du = dx$; $v = \int e^{-x} dx = -e^{-x}$

$\Rightarrow \int x e^{-x} dx = -x e^{-x} + \int e^{-x} dx$

$= -x e^{-x} + (-e^{-x})$

$= -x e^{-x} - e^{-x} = -e^{-x}(x+1)$

Hence $\int x^2 e^{-x} dx = -x^2 e^{-x} + 2[-e^{-x}(x+1)] + C$

$= -e^{-x}(x^2 + 2x + 2) + C$

Example 6

$\int e^x \sin x \, dx = ?$

Using integration by parts $\int u dv = uv - \int v du$

Let $u = \sin x$; $dv = e^x dx$

$du = \cos x dx$; $v = \int e^x dx = e^x$

$\Rightarrow \int e^x \sin x dx = e^x \sin x - \int e^x \cos x \, dx$ ——— (1)

We use integration by parts again for the latter integral:

Let $u = \cos x$; $dv = e^x dx$

$du = -\sin x dx$; $v = e^x$

$\Rightarrow \int e^x \cos x dx = e^x \cos x + \int e^x \sin x \, dx$ ——— (2)

Scanned by CamScanner

Substituting ② in ① :

$$\int e^x \sin x\,dx = e^x \sin x - (e^x \cos x + \int e^x \sin x\,dx)$$

$$\int e^x \sin x\,dx = e^x \sin x - e^x \cos x - \int e^x \sin x\,dx$$

Thus $2\int e^x \sin x\,dx = e^x(\sin x - \cos x) + C$

$$\Rightarrow \int e^x \sin x\,dx = \frac{1}{2} e^x(\sin x - \cos x) + C$$

Example 7

$$\int x \cos(5x-1) = ?$$

Let $u = x$; $dv = \cos(5x-1)\,dx$

$du = dx$; $v = \int \cos(5x-1)\,dx$

$$= \frac{1}{5}\sin(5x-1)$$

$$\Rightarrow \int x \cos(5x-1)\,dx = \frac{1}{5} x \sin(5x-1) - \frac{1}{5}\int \sin(5x-1)\,dx$$

$$= \frac{1}{5} x \sin(5x-1) + \left(\frac{1}{5}\right)\left(\frac{1}{5}\right)\cos(5x-1) + C$$

$$= \frac{1}{25}[5x \sin(5x-1) + \cos(5x-1)] + C$$

Try the following as exercises

Exx 1

$$\int x^3 e^x\,dx = ?$$

Let $u = x^3$, $dv = e^x dx$

$du = 3x^2$; $v = e^x$

$$\Rightarrow \int x^3 e^x dx = x^3 e^x - 3\int x^2 e^x dx \quad \text{——} ①$$

Scanned by CamScanner

Applying integration by parts on the latter integ

let $u = x^2$; $dv = e^x dx$

$du = 2x$; $v = e^x$

$\Rightarrow \int x^2 e^x dx = x^2 e^x - 2\int x e^x dx$ ——— (2)

Again, applying integration by parts on the latter integral

let $u = x$; $dv = e^x dx$

$du = dx$; $v = \int e^x dx = e^x$

Thus $\int x e^x dx = x e^x - \int e^x dx$

$= x e^x - e^x + C$

$= e^x(x-1)$ ——— (3)

Substituting (3) in (2) ;

$\int x^2 e^x dx = x^2 e^x - 2(e^x[x-1])$

$= x^2 e^x - 2e^x(x-1)$ ——— (4)

Substituting (4) in (1) :

$\int x^3 e^x dx = x^3 e^x - 3\left[x^2 e^x - 2e^x(x-1)\right] + C$

$= x^3 e^x - 3x^2 e^x + 6e^x(x-1) + C$

$= e^x(x^3 - 3x^2 + 6x - 6) + C$

Scanned by CamScanner

Ex 7

$$\int \frac{\ln 3x^2}{x^5}\,dx = ?$$

Let $u = \ln 3x^2$; $dv = \frac{1}{x^5}dx$

$du = \frac{1}{3x^2}\,6x\,dx$; $v = \int x^{-5}dx = \frac{x^{-4}}{-4}$

$du = \frac{6x\,dx}{3x^2} = \frac{2dx}{3}$; $v = -\frac{1}{4x^4}$

$$\Rightarrow \int \frac{\ln 3x^2}{x^5}dx = -\frac{1}{4x^4}\ln 3x^2 - \int -\frac{1}{4x^4}\cdot\frac{2}{x}dx$$

$$= -\frac{1}{4x^4}\ln 3x^2 - \frac{1}{2}\int x^{-5}dx$$

$$= -\frac{1}{4x^4}\ln 3x^2 + \left(\frac{1}{2}\right)\left(-\frac{1}{4}x^{-4}\right) + C$$

$$= -\frac{1}{8x^4}\left(2\ln 3x^2 + 1\right) + C$$

Scanned by CamScanner

Ex 8

$\int x(\ln 8x)^2 dx = ?$

Let $u = (\ln 8x)^2$; $dv = x\,dx$

$du = 2(\ln 8x)\cdot\frac{1}{8x}\cdot 8dx$ and $v = \int x dx$.

$\Rightarrow du = \frac{2}{x}\ln 8x\,dx$ and $v = \frac{1}{2}x^2$

Thus

$$\int x(\ln 8x)^2 dx = \frac{1}{2}x^2(\ln 8x)^2 - \int \frac{x^2}{2}\cdot\frac{2}{x}\ln 8x\,dx$$

$$= \frac{1}{2}x^2(\ln 8x)^2 - \int x\ln 8x\,dx \quad \text{——①}$$

For $\int x\ln 8x\,dx$,

Let $u = \ln 8x$ and $dv = x\,dx$

$du = \frac{1}{8x}\cdot 8dx$ and $v = \frac{1}{2}x^2$

$= \frac{1}{x}dx$

$$\Rightarrow \int x\ln 8x\,dx = \frac{1}{2}x^2\ln 8x - \int \frac{x^2}{2}\cdot\frac{1}{x}dx$$

$$= \frac{1}{2}x^2\ln 8x - \frac{1}{2}\int x\,dx$$

$$= \frac{1}{2}x^2\ln 8x - \frac{1}{2}\cdot\frac{1}{2}x^2$$

$$= \frac{1}{4}x^2(2\ln 8x - 1) \quad \text{——②}$$

Scanned by CamScanner

Putting ② in ① ;

$$\int x(\ln 8x)^2 dx = \frac{1}{2}x^2(\ln 8x)^2 - \frac{1}{4}x^2(2\ln 8x - 1) + C$$

$$= \frac{1}{4}x^2\left([2(\ln 8x)^2 - (2\ln 8x - 1)\right] + C$$

Ex 9

$$\int 9x^8 \ln 6x^5 dx = ?$$

Let $u = \ln 6x^5$ and $dv = 9x^8 dx$

$du = \frac{1}{6x^5} \cdot 30x^4 dx$ and $v = 9\int x^8 dx$

$= \frac{9 \cdot x^9}{9}$

$du = \frac{5}{x} dx$ and $v = x^9$

$$\Rightarrow \int 9x^8 \ln 6x^5 dx = x^9 \ln 6x^5 - \int x^9 \cdot \frac{5}{x} dx$$

$$= x^9 \ln 6x^5 - 5\int x^8 dx$$

$$= x^9 \ln 6x^5 - \frac{5}{9}x^9 + C$$

$$= x^9\left(\ln 6x^5 - \frac{5}{9}\right) + C$$

$$= \frac{1}{9}x^9(9\ln 6x^5 - 5) + C$$

Scanned by CamScanner

INVERSE TRIGONOMETRIC FUNCTIONS

Example 1

$$\int \sin^{-1}x\,dx = ?$$

Let $u = \sin^{-1}x$ and $dv = dx$

$$du = \frac{1}{\sqrt{1-x^2}}dx \text{ and } v = \int dx = x$$

$$\Rightarrow \int \sin^{-1}x\,dx = x\sin^{-1}x - \int \frac{x}{\sqrt{1-x^2}}dx \quad \text{——(1)}$$

Now we have $\int \frac{x}{\sqrt{1-x^2}}dx$ to integrate :

$$\int \frac{x}{\sqrt{1-x^2}}dx = ?$$

Let $u = 1-x^2$

$$du = -2x\,dx$$

$$\Rightarrow -\frac{du}{2} = x\,dx$$

$$\Rightarrow \int \frac{x}{\sqrt{1-x^2}}dx = \int \frac{1}{\sqrt{u}}\cdot\left(-\frac{du}{2}\right)$$

$$= -\frac{1}{2}\int u^{-\frac{1}{2}}du$$

$$= -\frac{1}{2}\cdot 2u^{\frac{1}{2}} = -\sqrt{u} = -\sqrt{1-x}$$

$$\Rightarrow \int \frac{x}{\sqrt{1-x^2}}dx = -\sqrt{1-x^2} \quad \text{——(2)}$$

Scanned by CamScanner

Substituting ② in ① we have

$$\int \sin^{-1}x\,dx = x\sin^{-1}x - (-\sqrt{1-x^2}) + c$$
$$= x\sin^{-1}x + \sqrt{1-x^2} + c$$

Example 2

$\int \cos^{-1}x\,dx = ?$

Let $u = \cos^{-1}x$ and $dv = dx$

$$du = -\frac{1}{\sqrt{1-x^2}}dx \quad \text{and} \quad v = \int dx = x$$

$$\Rightarrow \int \cos^{-1}x\,dx = x\cos^{-1}x - \int \left(-\frac{1}{\sqrt{1-x^2}}dx\right) x$$

$$\Rightarrow \int \cos^{-1}x\,dx = x\cos^{-1}x + \int \frac{x}{\sqrt{1-x^2}}dx \quad \text{——— ①}$$

We have to evaluate the latter integral

$$\int \frac{x}{\sqrt{1-x^2}}dx = ?$$

Let $u = 1-x^2$

$du = -2x\,dx$

$$\Rightarrow -\frac{du}{2} = x\,dx$$

$$\Rightarrow \int \frac{x}{\sqrt{1-x^2}}dx = \int \frac{1}{\sqrt{u}}\cdot\left(-\frac{du}{2}\right)$$

Scanned by CamScanner

$$= -\frac{1}{2}\int u^{-\frac{1}{2}}\,du$$

$$= -\frac{1}{2}\cdot 2u^{\frac{1}{2}} \quad = -\sqrt{u}$$

$$= -\sqrt{1-x^2}$$

$$\Rightarrow \int \frac{x}{\sqrt{1-x^2}}dx = -\sqrt{1-x^2} \qquad \text{——— (2)}$$

Putting (2) in (1) yields

$$\int \cos^{-1}x\,dx = x\cos^{-1}x + (-\sqrt{1-x^2}) + C$$

$$= x\cos^{-1}x - \sqrt{1-x^2} + C$$

Example 3

$$\int \tan^{-1}x\,dx = ?$$

Let $u = \tan^{-1}x$ and $dv = dx$

$$du = \frac{1}{1+x^2}dx \text{ and } v = \int dx = x$$

$$\Rightarrow \int \tan^{-1}x\,dx = x\tan^{-1}x - \int x\left(\frac{1}{1+x^2}dx\right.$$

$$= x\tan^{-1}x - \int \frac{x}{1+x^2}dx$$

$$= x\tan^{-1}x - \frac{1}{2}\int \frac{2x}{1+x^2}\,dx$$

Scanned by CamScanner

$$= x\tan^{-1}x - \frac{1}{2}\ln|1+x^2| + C$$

$$\Rightarrow \int \tan^{-1}x = x\tan^{-1}x - \frac{1}{2}\ln|1+x^2| + C$$

Example 4

$$\int x^2 \tan^{-1}x\,dx = ?$$

Let $u = \tan^{-1}x$; $dv = x^2dx$

$$du = \frac{1}{1+x^2}dx \; ; \; v = \int x^2dx = \frac{1}{3}x^3$$

$$\Rightarrow \int x^2\tan^{-1}x = \frac{1}{3}x^3\tan^{-1}x - \frac{1}{3}\int \frac{x^3}{1+x^2}dx$$

$$= \frac{1}{3}x^3\tan^{-1}x - \frac{1}{3}\int\left(x - \frac{x}{1+x^2}\right)dx$$

$$= \frac{1}{3}x^3\tan^{-1}x - \frac{1}{3}\left[\frac{1}{2}x^2 - \frac{1}{2}\ln|1+x^2|\right] + C$$

$$= \frac{1}{3}x^3\tan^{-1}x - \frac{1}{6}x^2 + \frac{1}{6}\ln|1+x^2| + C$$

Example 5

$$\int (\sin^{-1}x)^2 dx = ?$$

Let $u = \sin^{-1}x \Rightarrow x = \sin u$

$$du = \frac{1}{\sqrt{1-x^2}}dx$$

Scanned by CamScanner

$$\Rightarrow dx = \sqrt{1-x^2}\,du$$

$$\Rightarrow \int (\sin^{-1}x)^2 dx = \int u^2\sqrt{1-x^2}\,du$$

$$= \int u^2\sqrt{1-\sin^2 u}\,du$$

$$= \int u^2 \cos u\,du$$

Applying integration by parts to the integral

$$\int u^2 \cos u\,du = ?$$

TRY IT YOURSELF

$$\int u^2 \cos u\,du = (u^2-2)\sin u + 2u\cos u) + C$$

$$= [(\sin^{-1}x)^2 - 2]x + 2(\sin^{-1}x)\sqrt{1-x^2} + C$$

$$\therefore \int (\sin^{-1}x)^2 dx = [(\sin^{-1}x)^2 - 2]x + 2(\sin^{-1}x)\sqrt{1-x^2} + C$$

Exercises

1 (i) $\int x \sin x\,dx$ (ii) $\int x \sin 3x\,dx$

(iii) $\int x^2 \cos x$ (iii) $\int x \ln x\,dx$

2. (i) $\int x e^{-ax}\,dx$ (ii) $\int \tan^{-1}x\,dx$

(iii) $\int x \tan^{-1}x\,dx$ (iv) $\int \cos^{-1}x\,dx$

Scanned by CamScanner

Exx 2

$\int e^{ax} \cos bx \, dx = ?$

Let $u = \cos bx$ and $dv = e^{ax} dx$

$du = -b \sin bx \, dx$ and $v = \frac{1}{a} e^{ax}$

$\Rightarrow \int e^{ax} \cos bx \, dx = \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} \int e^{ax} \sin bx \, dx$

Applying integration by parts to the latter

Let $u = \sin bx$; $dv = e^{ax} dx$

$du = b \cos bx$; $v = \frac{1}{a} e^{ax}$

$\Rightarrow \int e^{ax} \sin bx \, dx = \frac{1}{a} e^{ax} \sin bx - \frac{b}{a} \int e^{ax} \cos bx \, dx$

Substituting ② in ① yields

$\int e^{ax} \cos bx \, dx = \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} \left[\frac{1}{a} e^{ax} \sin bx - \frac{b}{a} \int e^{ax} \cos \right.$

$= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{ax} \sin bx - \frac{b^2}{a^2} \int e^{ax} \cos$

$\Rightarrow \int e^{ax} \cos bx \, dx + \frac{b^2}{a^2} \int e^{ax} \cos x \, dx = \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{a}$

$\left(1 + \frac{b^2}{a^2}\right) \int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2} (a \cos bx + b \sin bx$

Scanned by CamScanner

$$\Rightarrow \int e^{ax} \cos bx\, dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$$

Ex 3

$$\int e^{ax} \cos(bx+c)\, dx = ?$$

Let $u = \cos(bx+c)$ and $dv = e^{ax} dx$

$du = -b\sin(bx+c)$ and $v = \frac{1}{a} e^{ax}$

$$\int e^{ax} \cos(bx+c)\, dx = \frac{1}{a} e^{ax} \cos(bx+c) + \frac{b}{a}\int e^{ax} \sin(bx+c)\, dx$$

Applying integration by parts to the integral $\int e^{ax} \sin(bx+c)\, dx$, we have

$$\int e^{ax} \sin(bx+c)\, dx = \frac{1}{a} e^{ax} \sin(bx+c) - \frac{b}{a}\int e^{ax} \cos(bx+c)\, dx$$

$$\Rightarrow \int e^{ax} \cos(bx+c)\, dx = \frac{1}{a} e^{ax} \cos(bx+c) + \frac{b}{a}\left(\frac{1}{a} e^{ax} \sin(bx+c) - \frac{b}{a}\int e^{ax} \cos(bx+c)\, dx\right)$$

$$= \frac{1}{a} e^{ax} \cos(bx+c) + \frac{b}{a^2} e^{ax} \sin(bx+c) - \frac{b^2}{a^2}\int e^{ax} \cos(bx+c)\, dx$$

Scanned by CamScanner

$$= \frac{e^{ax}}{a^2+b^2}\left[a\cos(bx+c)+b\sin(bx+c)\right]+D$$

$$\Rightarrow \int e^{ax}\cos(bx+c)\,dx = \frac{e^{ax}}{a^2+b^2}\left[a\cos(bx+c)+b\sin(bx+c)\right]+D$$

STANDARD INTEGRALS OF SOME FUNCTIONS

1. $\int e^{ax}\sin bx\,dx = \dfrac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$

2. $\int e^{ax}\cos bx\,dx = \dfrac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$

Exx 4

$\int x^2 e^{3x}\,dx$

Let $y = 3x$ and $dy = 3dx$ $\left(dx = \frac{dy}{3}\right)$

$\Rightarrow x = \frac{y}{3}$; $x^2 = \frac{y^2}{9}$

Thus $\int x^2 e^{3x}\,dx = \int \frac{y^2}{9}\cdot e^y \frac{dy}{3}$

$= \frac{1}{27}\int y^2 e^y\,dy$

Scanned by CamScanner

Let $u = y^2$; $\frac{dv}{dy} = e^y$

$\Rightarrow \frac{du}{dy} = 2y$; $v = e^y$

$$\frac{1}{27}\int y^2 e^y \, dy = ?$$

$$= \frac{1}{27}\left(y^2 e^y - 2\int y e^y \, dy\right) \text{---} \circledast$$

To calculate the latter, we use another integration by parts ;

$$2\int y\, e^y \, dy = ?$$

Let $u = y$; $\frac{dv}{dy} = e^y$

$\frac{du}{dy} = 1$; $v = e^y$

$$2\int y\, e^y \, dy = 2\left[y e^y - \int e^y \, dy\right]$$

$$= 2\left[y e^y - e^y\right]$$

$$= 2y e^y - 2e^y$$

Scanned by CamScanner

Let $x = -3$ Then $1 = -6B$, $B = -\frac{1}{6}$

So $\frac{1}{x^2-9} = \frac{1}{6}\left[\frac{1}{x-3}\right] - \frac{1}{6}\left[\frac{1}{x+3}\right]$

Hence $\int \frac{dx}{x^2-9} = \frac{1}{6}\ln|x-3| - \frac{1}{6}\ln|x+3| + c$

$= \frac{1}{6}\ln\left|\frac{x-3}{x+3}\right| + c$

Example 2

$\int \frac{x\,dx}{(x+2)(x+3)} = ?$

$\frac{x}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3}$

Then $x = A(x+3) + B(x+2)$

Let $x = -3$ Then $-3 = -B$, $B = 3$

Let $x = -2$ Then $-2 = A$

So $\frac{x}{(x+2)(x+3)} = -\frac{2}{x+2} + \frac{3}{x+3}$

Hence $\int \frac{x\,dx}{(x+2)(x+3)} = -2\ln|x+2| + 3\ln|x+3| + c$

$= \ln\left|\frac{(x+3)^3}{(x+2)^2}\right| + c$

Scanned by CamScanner

Example 3

$$\int \frac{x^4 - 4x^2 + x + 1}{x^2 - 4}\,dx = ?$$

Since the degree of the numerator is at least as great as that of the denominator, carry out the long division,

obtaining $\frac{x^4 - 4x^2 + x + 1}{x^2 - 4} = x^2 + \frac{x+1}{x^2-4}$

Thus $\int \frac{x^4 - 4x^2 + x + 1}{x^2 - 4}\,dx = \frac{1}{3}x^3 + \int \frac{x+1}{x^2-4}\,dx$

But $\frac{x+1}{x^2-4} = \frac{x+1}{(x+2)(x-2)} = \frac{A}{x+2} + \frac{B}{x-2}$

Then $x + 1 = A(x-2) + B(x+2)$

Let $x = 2$. Then $3 = 4B$, $B = \frac{3}{4}$

Let $x = -2$. Then $-1 = -4A$, $A = \frac{1}{4}$

Thus, $\frac{x+1}{(x+2)(x-2)} = \frac{1}{4}\cdot\frac{1}{x+2} + \frac{3}{4}\cdot\frac{1}{x-2}$,

and $\int \frac{x+1}{(x+2)(x-2)}\,dx = \frac{1}{4}\ln|x+2| + \frac{3}{4}\ln|x-2|$

$= \frac{1}{4}\ln\left|\frac{x+2}{(x-2)^3}\right|$

Scanned by CamScanner

Hence, the complete answer is

$$\frac{1}{3}x^3 + \frac{1}{4}\ln\left|\frac{x+2}{(x-2)^3}\right| + C$$

Example 4

$$\int \frac{2x^2+1}{(x-1)(x-2)(x-3)}\,dx = ?$$

$$\frac{2x^2+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

Then $2x^2+1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$

Let $x=1$. Then $3 = 2A$, $A = \frac{3}{2}$.

Let $x=2$. Then $9 = -B$, $B = -9$.

Let $x=3$. Then $19 = 2C$, $C = \frac{19}{2}$

Thus $\frac{2x^2+1}{(x-1)(x-2)(x+3)} = \frac{3}{2}\frac{1}{x-1} - \frac{9}{x-2} + \frac{19}{2}\frac{1}{x-3}$

Hence,

$$\int \frac{2x^2+1}{(x-1)(x-2)(x-3)}\,dx = \frac{3}{2}\ln|x-1| - 9\ln|x-2| + \frac{19}{2}\ln|x-3| + C_1$$

$$= \frac{1}{2}\ln\left|\frac{(x-1)^3(x-3)^{19}}{(x-2)^{18}}\right| + C_1$$

Scanned by CamScanner

Example 5

$$\int \frac{x^2-4}{x^3-3x^2-x+3}\,dx = ?$$

Now $\dfrac{x^2-4}{x^3-3x^2-x+3} = \dfrac{x^2-4}{(x-1)(x-3)(x+1)} = \dfrac{A}{x-1} + \dfrac{B}{x-3} + \dfrac{C}{x+1}$

$\Rightarrow x^2-4 = A(x-3)(x+1) + B(x-1)(x+1) + C(x-1)(x-3$

Let $x=1, \Rightarrow -3 = -4A \Rightarrow A = \frac{3}{4}$

Let $x=3 \Rightarrow 5 = 8B \Rightarrow B = \frac{5}{8}$

Let $x=-1 \Rightarrow -3 = 8C \Rightarrow C = \frac{-3}{8}$

$$\Rightarrow \frac{x^2-4}{(x-1)(x-3)(x+1)} = \frac{3}{4}\left(\frac{1}{x-1}\right) + \frac{5}{8}\left(\frac{1}{x-3}\right) - \frac{3}{8}\left(\frac{1}{x+1}\right)$$

$$\Rightarrow \int \frac{x^2-4}{(x-1)(x-3)(x+1)}\,dx = \frac{3}{4}\ln|x-1| + \frac{5}{8}\ln|x-3| - \frac{3}{8}\ln|x+1| +$$

$$= \frac{1}{8}(6\ln|x-1| + 5\ln|x-1| - 3\ln|x+1| +$$

$$= \frac{1}{8}\ln\left|\frac{(x-1)^6(x-3)^5}{(x+1)^3}\right| + C$$

$$\Rightarrow \int \frac{x^2-4}{(x-1)(x-3)(x+1)}\,dx = \frac{1}{8}\ln\left|\frac{(x-1)^6(x-3)^5}{(x+1)^3}\right| + C$$

Scanned by CamScanner

Example 66

$$\int \frac{x\,dx}{x^4-13x^2+36} = ?$$

$$x^4-13x^2+36 = (x^2-9)(x^2-4)$$
$$= (x-3)(x+3)(x+2)(x-2)$$

Let $\frac{x}{(x-3)(x+3)(x+2)(x-2)} = \frac{A}{x-3}+\frac{B}{x+3}+\frac{C}{x+2}+\frac{D}{x-2}$

$$\Rightarrow x = A(x+3)(x+2)+B(x-3)(x+2)(x-2)+C(x-3)(x+3)(x-2)$$
$$+ D(x-3)(x+3)(x+2).$$

Let $x = 3 \Rightarrow 3 = 30A,\ A = \frac{1}{10}$

Let $x = -3 \Rightarrow -3 = -30B,\ B = \frac{1}{10}$

Let $x = -2 \Rightarrow -2 = 20C,\ C = -\frac{1}{10}$

Let $x = 2 \Rightarrow 2 = -20D,\ D = -\frac{1}{10}$

$$\frac{x}{x^4-13x^2+36} = \frac{1}{10}\frac{1}{x-3}+\frac{1}{10}\frac{1}{x+3}-\frac{1}{10}\frac{1}{x+2}-\frac{1}{10}\frac{1}{x-2}$$

and $$\int \frac{x}{x^4-13x^2+36}\,dx = \frac{1}{10}\ln|x-3|+\frac{1}{10}\ln|x+3|-\frac{1}{10}\ln|x+2|$$
$$-\frac{1}{10}\ln|x-2|+C_1$$
$$= \frac{1}{10}(\ln|x-3|+\ln|x+3|-\ln|x+2|$$
$$-\ln|x+2|)+C_1$$

Scanned by CamScanner

Scanned by CamScanner

To be treated as exercises

Exx 1

$$\int \frac{x^4}{x^3-2x^2-7x-4}\,dx = ?$$

Now

$$\frac{x^4}{x^3-2x^2-7x-4} = ?$$

First we divide the numerator by the denominator obtaining $x+2+\frac{11x^2+18x+8}{x^3-2x^2-7x-4}$

$$\Rightarrow \int \frac{x^4}{x^3-2x^2-7x-4}\,dx = \int\left\{x+2+\frac{11x^2+18x+8}{x^3-2x^2-7x-4}\right\}dx$$

$$= \int (x+2)\,dx + \int \frac{11x^2+18x+8}{x^3-2x^2-7x-4}\,dx$$

$$= \frac{1}{2}x^2+2x + \int \frac{11x^2+18x+8}{x^3-2x^2-7x-4}\,dx$$

Thus we have to evaluate the latter integral

$$\int \frac{11x^2+18x+8}{x^3-2x^2-7x-4}\,dx = ?$$

$$\frac{11x^2+18x+8}{x^3-2x^2-7x-4} = ?$$

Here we seek to factorise the denominator. Looking for its roots, we examine the integral factors of the constant term, -4.

Scanned by CamScanner

We find that $x=-1$ is a root of the denominator; dividing x^3-2x^2-7x-4 by $x+1$ yields

$$x^2-3x-4 = (x-4)(x+1)$$

$\Rightarrow$ denominator is $(x-4)(x+1)^2$

$$\Rightarrow \frac{11x^2+18x+8}{(x-4)(x+1)^2} = \frac{A}{x-4} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

$$\Rightarrow 11x^2+18x+8 = A(x+1)^2 + B(x-4)(x+1) + C(x-4)$$

Let $x=4 \Rightarrow 256 = 25A \Rightarrow A = \frac{256}{25}$

Let $x=-1 \Rightarrow 1 = -5C \Rightarrow C = -\frac{1}{5}$

To find B, equate coefficients of x^2:

$$11 = A + B \Rightarrow B = 11 - A = \frac{19}{25}$$

Thus

$$\frac{11x^2+18x+8}{(x-4)(x+1)^2} = \frac{256}{25}\left(\frac{1}{x-4}\right) + \frac{19}{25}\left(\frac{1}{x+1}\right) - \frac{1}{5}\left(\frac{1}{(x+1)^2}\right)$$

$$\Rightarrow \int \frac{11x^2+18x+8}{(x-4)(x+1)^2}dx = \int\left[\frac{256}{25}\left(\frac{1}{x-4}\right) + \frac{19}{25}\left(\frac{1}{x+1}\right) - \frac{1}{5}\left(\frac{1}{(x+1)^2}\right)\right]dx$$

$$= \frac{256}{25}\ln|x-4| + \frac{19}{25}\ln|x+1| + \frac{1}{5}\frac{1}{(x+1)} + C$$

Finally we have

$$\frac{x^4\,dx}{x^3-2x^2-7x-4} = \frac{1}{2}x^2 + 2x + \frac{256}{25}\ln|x-4| + \frac{19}{25}\ln|x+1|$$

Scanned by CamScanner

Ex 5

$$\int \frac{dx}{x(x^2+1)^2} = ?$$

$$\frac{1}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

$$\Rightarrow 1 = A(x^2+1)^2 + (Bx+C)x(x^2+1) + (Dx+E)x$$

Let $x = 0 \Rightarrow 1 = A$

Equating coefficients of x^4: $0 = A + B \Rightarrow B = -A$

$\Rightarrow B = -1$

Equating coefficients of x^3: $0 = C$

Equating coefficients of x^2: $0 = 2A + B + D$

$\Rightarrow D = -2A - B = -1$

Equating coefficients of x: $0 = C + E$

$\Rightarrow E = -C = 0$

$$\frac{1}{x(x^2+1)^2} = \frac{1}{x} - \frac{1}{x^2+1} - \frac{1}{(x^2+1)^2}$$

Now

$$\int \frac{1}{x(x^2+1)^2}\,dx = \int \left[\frac{1}{x} - \frac{1}{x^2+1} - \frac{1}{(x^2+1)^2}\right] dx$$

Scanned by CamScanner

$$= \int \frac{1}{x}dx - \int \frac{1}{x^2+1}dx - \int \frac{1}{(x^2+1)^2}dx$$

$$= \ln|x| - \tan^{-1}x - \left[\frac{1}{2}\tan^{-1}x + \frac{x}{x^2+1}\right] + C$$

$$= \ln|x| - \frac{3}{2}\tan^{-1}x - \frac{1}{2}\left(\frac{x}{x^2+1}\right) + C$$

Try and check the solution for

$$\int \frac{1}{(x^2+1)^2}dx = \left[\frac{1}{2}\tan^{-1}x + \frac{x}{x^2+1}\right] + C$$

(check ↓

Let $x = \tan\theta$

$dx = \sec^2\theta$)

Ex 6

$$\int \frac{x^2dx}{(x-1)(x^2+4)^2} = ?$$

$$\frac{x^2}{(x-1)(x^2+4)^2} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4} + \frac{Dx+E}{(x^2+4)^2}$$

$$\Rightarrow x^2 = A(x^2+4)^2 + (Bx+C)(x-1)(x^2+4) + (Dx+E)(x-1)$$

Let $x = 1 \Rightarrow 1 = 25A \Rightarrow A = \frac{1}{25}$

Equating coefficients of x^4

$0 = A + B \Rightarrow B = -A = -\frac{1}{25}$

Scanned by CamScanner

Equating coefficients of x^3:

$$0 = -B + C \Rightarrow C = B = -\frac{1}{25}$$

Equating coefficients of x^2:

$$1 = 8A + 4B - C + D$$

$$\Rightarrow D = \frac{20}{25} = \frac{4}{5}$$

Equating constant terms:

$$0 = 16A - 4C - E$$

$$\Rightarrow E = 16A - 4C$$

$$= \frac{20}{25} = \frac{4}{5}$$

Thus $$\frac{x^2}{(x-1)(x^2+4)^2} = \frac{1}{25}\left(\frac{1}{x-1}\right) - \frac{1}{25}\left(\frac{x+1}{x^2+4}\right) + \frac{4}{5}\left(\frac{x+1}{[x^2+4]^2}\right)$$

$$= \int\left[\frac{1}{25}\left(\frac{1}{x-1}\right) - \frac{1}{25}\left(\frac{x+1}{x^2+4}\right) + \frac{4}{5}\left(\frac{x+1}{(x+4)^2}\right)\right]dx$$

$$= \frac{1}{25}\ln|x-1| - \frac{1}{25}\int\frac{x+1}{x^2+4}dx + \frac{4}{5}\int\frac{x+1}{(x+4)^2}dx$$

Now

$$\int\frac{x+1}{x^2+4}dx = \int\frac{x\,dx}{x^2+4} + \int\frac{dx}{x^2+4} = \frac{1}{2}\ln(x^2+4) + \frac{1}{2}\tan^{-1}\frac{x}{2}$$

(Let $x = 2\tan\theta$, $dx = 2\sec^2\theta$)

Scanned by CamScanner

Also

$$\int \frac{(x+1)dx}{(x^2+4)^2} = \int \frac{x\,dx}{(x^2+4)^2} + \int \frac{dx}{(x^2+4)^2}$$

$$= \frac{1}{2}\left(\frac{1}{x^2+4}\right) + \frac{1}{16}\left(\tan^{-1}\frac{x}{2} + \frac{2x}{x^2+4}\right)$$

The complete solution is given as

$$\int \frac{x^2}{(x-1)(x^2+4)^2}dx = \frac{1}{25}\ln|x-1| - \frac{1}{25}\left[\frac{1}{2}\ln|x^2+4| + \frac{1}{2}\tan^{-1}\frac{x}{2}\right]$$

$$+ \frac{4}{5}\left(\frac{1}{8}\left[\frac{x-4}{x^2+4}\right] + \frac{1}{16}\tan^{-1}\frac{x}{2}\right) + C$$

$$= \frac{1}{25}\ln\frac{|x-1|}{\sqrt{x^2+4}} + \frac{3}{100}\tan^{-1}\frac{x}{2} + \frac{1}{10}\left(\frac{x-4}{x^2+4}\right) + C$$

Note that when the denominator of the algebraic fraction is the sum of two squares, we cannot factorise it and had to resort to a different strategy for integration.

The method is to use two standard forms of integration as below:

Scanned by CamScanner

DINARY DIFFERENTIAL EQUATIONS

many physical situations, equations arise ich involve differential coefficients. For ample:

1. The rate of decrease of temperature of a hot body
2. A body falling freely under gravity
3. The oscillation on the end of a spring
4. The decay of a radioactive substance
5. The decrease in concentration of chemical compound in a reaction

ORMATION OF DIFFERENTIAL EQUATIONS

ifferential equations arise or may be derived a variety of ways. In most cases the problem to find the dependent variable in terms of the independent one. Eg. Finding x in terms of t or y in terms of x.

Differential equations may be formed by

Scanned by CamScanner

direct differentiation

Eg $y = x^3 + 7x^2 + 3x + 7$ ——— (1)

$\frac{dy}{dx} = 3x^2 + 14x + 3$ ——— (2)

$\frac{d^2y}{dx^2} = 6x + 14$ ——— (3)

$\frac{d^3x}{dx^3} = 6$ ——— (4)

Defn:

Equations which contain an independent variable a dependent variable and at least one of their derivatives are called DIFFERENTIAL EQUATIONS

Thus a relationship between a variable quantity x and a dependent function y and its derivatives $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$ ..... is called an ORDINARY DIFFERENTIAL EQUATION

Equations (2), (3) and (4) are differential equations

Scanned by CamScanner

er examples of ordinary differential equations

$$\frac{dy}{dx} = kx$$

$$x^2(1+y)\frac{dy}{dx} - (1+x)y^2 = 0$$

$$\frac{d^2y}{dx^2} = -n^2y$$

$$\left(\frac{dy}{dx}\right)^2 + 3y = 0$$

$$x\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^4 = 0$$

KINDS OF DIFFERENTIAL EQUATIONS

e are two main types of differential quations

ORDINARY DIFFERENTIAL EQUATIONS (O.D.E's)

These are differential equations with only one independent variable.

PARTIAL DIFFERENTIAL EQUATIONS

These are differential equations with more than one independent variable.

Scanned by CamScanner

Examples of partial differential equations

1. $x\frac{\partial z}{\partial y} + y\frac{\partial z}{\partial x} = 0$

2. $\frac{\partial^2 f}{\partial y^2} = \frac{2x}{x^2+y^2}$

3. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$

4. $f_{xx} + f_{yy} = 0$

5. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial z}{\partial t}$

NOTE:
This course is concerned more with ordinary differential equations. Partial differential equations may be considered slightly

ORDER OF A DIFFERENTIAL EQUATION

Differential equations are classified according to the highest derivative which occurs in the equations.
Thus if the highest derivative that occurs in an equation is $\frac{d^n y}{dx^n}$, the equation is said

Scanned by CamScanner

be of order n

Considering the examples in page 154, we notice that Examples ①, ② and ④ are of first order types. Example 3 is of a second order type whilst example 5 is a third order ordinary differential equation.

In page 166, we also notice that

Equation 2 is of the first order, having only the first derivative

Equation 3 is of the second order

Equation 4 is of the third order

DEGREE OF A DIFFERENTIAL EQUATION

The degree of a differential equation is the highest powers of the highest derivative which the equation contains

Thus $\left(\frac{d^2y}{dx^2}\right)^3 + 3\frac{dy}{dx} = 0$ is of the second order and third degree

Considering the examples in page 167, we also

Scanned by CamScanner

notice that Equations ①, ②, ③ and ⑤ are of first degree whilst equation ④ is of a second degree

Note that in the example ⑤ the degree of the equation is determined by the power of the highest derivative $\frac{d^3y}{dx^3}$ and not by the fourth power term in $\frac{dy}{dx}$

Exx.

Indicate the order and the degree of the following ordinary differential equations

1. $\frac{dy}{dx} - 2xy = 2x$ O_1, d_1

2. $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} = y - x$ O_2, d_3

3. $\left(\frac{dy}{dx}\right)^4 + xy = x$ O_1, d_4

4. $x^2\frac{dy}{dx} + x\frac{d^4y}{dx^4} = 3y$ O_4, d_2

5. $e^x\frac{dy}{dx} = (1 - ye^x)$ O_1, d_1

Scanned by CamScanner